Sr. No.	Title of Experiment	Page No.	Date	Sign
	ML			
1	**Feature Transformation**: To use PCA Algorithm for dimensionality reduction. You have a dataset that includes measurements for different variables on wine (alcohol, ash, magnesium, and so on). Apply PCA algorithm & transform this data so that most variations in the measurements of the variables are captured by a small number of principal components so that it is easier to distinguish between red and white wine by inspecting these principal components.			
2	**Regression Analysis**: Use the diabetes data set from UCI and Pima Indians Diabetes data set for performing the following: a. Univariate analysis: Frequency, Mean, Median, Mode, Variance, Standard Deviation, Skewness and Kurtosis b. Bivariate analysis: Linear and logistic regression modeling c. Multiple Regression analysis d. Also compare the results of the above analysis for the two data sets.			
3	**Classification Analysis**: Implementation of Support Vector Machines (SVM) for classifying images of handwritten digits into their respective numerical classes (0 to 9).			
4	**Clustering Analysis**: Implement K-Means clustering on Iris.csv dataset. Determine the number of clusters using the elbow method.			
5	**Ensemble Learning**: Use different voting mechanism and Apply AdaBoost (Adaptive Boosting), Gradient Tree Boosting (GBM), XGBoost classification on Iris dataset and compare the performance of three models using different evaluation measures.			
6	**Reinforcement Learning**: Building the reinforcement learning model d. Training the model e. Testing the model.			
	DMV			
7	**Interacting with Web APIs** Problem Statement: Analyzing Weather Data from OpenWeatherMap API Dataset: Weather data retrieved from OpenWeatherMap API Description: The goal is to interact with the OpenWeatherMap API to retrieve weather data for a specific location and perform data modeling and visualization to analyze weather patterns over time. Tasks to Perform: 1. Register and obtain API key from OpenWeatherMap. 2. Interact with the OpenWeatherMap API using the API key to retrieve weather data for a specific location.			

	3. Extract relevant weather attributes such as temperature, humidity, wind speed, and precipitation from the API response. 4. Clean and preprocess the retrieved data, handling missing values or inconsistent formats. 5. Perform data modeling to analyze weather patterns, such as calculating average temperature, maximum/minimum values, or trends over time. 6. Visualize the weather data using appropriate plots, such as line charts, bar plots, or scatter plots, to represent temperature changes, precipitation levels, or wind speed variations. 7. Apply data aggregation techniques to summarize weather statistics by specific time periods (e.g., daily, monthly, seasonal). 8. Incorporate geographical information, if available, to create maps or geospatial visualizations representing weather patterns across different locations. 9. Explore and visualize relationships between weather attributes, such as temperature and humidity, using correlation plots or heatmaps.			
8	**Data Cleaning and Preparation** Problem Statement: Analyzing Customer Churn in a Telecommunications Company Dataset: "Telecom_Customer_Churn.csv" Description: The dataset contains information about customers of a telecommunications company and whether they have churned (i.e., discontinued their services). The dataset includes various attributes of the customers, such as their demographics, usage patterns, and account information. The goal is to perform data cleaning and preparation to gain insights into the factors that contribute to customer churn. Tasks to Perform: 1. Import the "Telecom_Customer_Churn.csv" dataset. 2. Explore the dataset to understand its structure and content. 3. Handle missing values in the dataset, deciding on an appropriate strategy. 4. Remove any duplicate records from the dataset. 5. Check for inconsistent data, such as inconsistent formatting or spelling variations, and standardize it. 6. Convert columns to the correct data types as needed. 7. Identify and handle outliers in the data. 8. Perform feature engineering, creating new features that may be relevant to predicting customer churn. 9. Normalize or scale the data if necessary. 10. Split the dataset into training and testing sets for further analysis. 11. Export the cleaned dataset for future analysis or modeling.			

9	**Data Wrangling** Problem Statement: Data Wrangling on Real Estate Market Dataset: "RealEstate_Prices.csv" Description: The dataset contains information about housing prices in a specific real estate market. It includes various attributes such as property characteristics, location, sale prices, and other relevant features. The goal is to perform data wrangling to gain insights into the factors influencing housing prices and prepare the dataset for further analysis or modeling. Tasks to Perform: 1. Import the "RealEstate_Prices.csv" dataset. Clean column names by removing spaces, special characters, or renaming them for clarity. 2. Handle missing values in the dataset, deciding on an appropriate strategy (e.g., imputation or removal). 3. Perform data merging if additional datasets with relevant information are available (e.g., neighborhood demographics or nearby amenities). 4. Filter and subset the data based on specific criteria, such as a particular time period, property type, or location. 5. Handle categorical variables by encoding them appropriately (e.g., one-hot encoding or label encoding) for further analysis. 6. Aggregate the data to calculate summary statistics or derived metrics such as average sale prices by neighborhood or property type. 7. Identify and handle outliers or extreme values in the data that may affect the analysis or modeling process			
10	**Data Visualization using matplotlib** Problem Statement: Analyzing Air Quality Index (AQI) Trends in a City Dataset: "City_Air_Quality.csv" Description: The dataset contains information about air quality measurements in a specific city over a period of time. It includes attributes such as date, time, pollutant levels (e.g., PM2.5, PM10, CO), and the Air Quality Index (AQI) values. The goal is to use the matplotlib library to create visualizations that effectively represent the AQI trends and patterns for different pollutants in the city. Tasks to Perform: 1. Import the "City_Air_Quality.csv" dataset. 2. Explore the dataset to understand its structure and content. 3. Identify the relevant variables for visualizing AQI trends, such as date, pollutant levels, and AQI values. 4. Create line plots or time series plots to visualize the overall AQI trend over time. 5. Plot individual pollutant levels (e.g., PM2.5, PM10, CO) on separate line plots to visualize their trends over time. Use bar plots or stacked bar plots to compare the AQI values across different dates or time periods. 7. Create box plots or violin plots to analyze the			

	distribution of AQI values for different pollutant categories. 8. Use scatter plots or bubble charts to explore the relationship between AQI values and pollutant levels. 9. Customize the visualizations by adding labels, titles, legends, and appropriate color schemas			
11	**Data Aggregation** Problem Statement: Analyzing Sales Performance by Region in a Retail Company Dataset: "Retail_Sales_Data.csv" Description: The dataset contains information about sales transactions in a retail company. It includes attributes such as transaction date, product category, quantity sold, and sales amount. The goal is to perform data aggregation to analyze the sales performance by region and identify the top-performing regions. Tasks to Perform: 1. Import the "Retail_Sales_Data.csv" dataset. 2. Explore the dataset to understand its structure and content. 3. Identify the relevant variables for aggregating sales data, such as region, sales amount, and product category. 4. Group the sales data by region and calculate the total sales amount for each region. 5. Create bar plots or pie charts to visualize the sales distribution by region. 6. Identify the top-performing regions based on the highest sales amount. 7. Group the sales data by region and product category to calculate the total sales amount for each combination. 8. Create stacked bar plots or grouped bar plots to compare the sales amounts across different regions and product categories.			
	Mini Project			
12	Mini Project: Development of a School Happiness and Well-being Index Using Machine Learning			

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	BI			
1	**DNA Sequence Analysis:** Analyze a given DNA sequence and perform basic sequence manipulation, including finding motifs, calculating GC content, and identifying coding regions			
2	**RNA-Seq Data Analysis:** Analyze a provided RNA-Seq dataset and perform differential gene expression analysis.			
3	**Protein Structure Prediction:** Predict the 3D structure of a given protein sequence using homology modeling or threading techniques.			
4	**Molecular Docking and Virtual Screening:** Perform molecular docking simulations to predict the binding affinity between a protein target and a small molecule ligand. Additionally, conduct virtual screening to identify potential drug candidates.			
5	**Machine Learning for Genomic Data**: Apply machine learning algorithms, such as random forests or support vector machines, to classify genomic data based on specific features or markers.			
6	**Agricultural Genomics and Crop Improvement**: Analyze genomic data from crops to identify genetic markers associated with desirable traits, such as disease resistance or yield.			
	IR			
7	Write a program for pre-processing of a text document such as stop word removal, stemming.			
8	Implement a program for retrieval of documents using inverted files.			
9	Write a program to construct a Bayesian network considering medical data. Use this model to demonstrate the diagnosis of heart patients using the standard Heart Disease Data Set (You can use Java/Python ML library classes/API.			
10	Implement Agglomerative hierarchical clustering algorithm using appropriate dataset.			
11	Implement Page Rank Algorithm. (Use python or beautiful soup for implementation).			
	Mini Project			
12	Mini Project: Differential Gene Expression Analysis of Disease and Normal Samples using Statistical Bioinformatics			